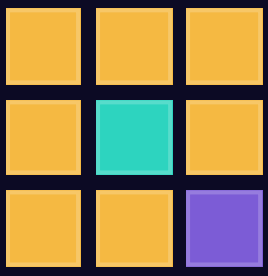
SQUEEKCHAIN GENESIS · WHITEPAPER V1.0

SQUEEK TILES

The square-grid elimination game where every player is a hamster tile, the wheel decides who falls, and the final tile was sealed by a block before the first spin.



A 3 x 3 board. Nine tiles in, one tile out. That is the whole promise - and everything in this paper exists to make that promise verifiable.

Network: Squeekchain - entry token SQK on Sepolia (test token, no real-world value)
Contract: SqueekTiles - **deployed on Sepolia** at `0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212`
SQK token: `0xd78a3956cF63164d1Fa1C622ad3c66623162Fa1e` (18 decimals)
Creator (fee recipient): `0x8a53644c0CF45EFc0247E8e01d24607b870793C4`
Playable now: Real Mode on Sepolia with MetaMask, plus Practice mode with identical rules, fake SQK and bot players - no wallet required
Live game site: To be added at submission (project domain pending)
Lobbies: 4 players (2 x 2) · 9 players (3 x 3) · 16 players (4 x 4)
Entry: 20 SQK = 18 SQK to the prize + 2 SQK creator fee

START HERE

Contents

1. Why Squeek Tiles exists
2. How it uses Squeekchain
3. The square rule and the three lobbies
4. The full process, step by step
5. Tokenomics - where every SQK goes
6. Worked examples with exact numbers
7. The raise system
8. The final tile: FULL / HALF / EMPTY
9. Treasury and the 6-hour airdrop
10. Practice mode vs on-chain mode
11. Smart contract specification
12. Fairness, security and honest limitations
13. Deployment record and submission
14. Timing: the final settings

Appendix: Glossary

START HERE

Executive summary

Squeek Tiles is an elimination game for a square board. Players join a lobby of exactly 4, 9 or 16. Each player becomes one hamster tile and pays a 20 SQK entry. Of that entry, 18 SQK is locked into the prize held in the final tile and 2 SQK is paid to the game creator as a fee. Once the lobby is full, the game starts and cannot be joined, paused or edited.

Each round, the hamster wheel spins for 10 seconds. There is no skip: the spin always runs its full 10 seconds for every player. When it stops, one address is drawn at random from the addresses still alive and that tile is eliminated. Rounds repeat until exactly one tile remains. That survivor reaches the final tile - but the final tile does not always pay in full. Its tier was fixed at the moment the game started, by a published formula applied to the Squeekchain block produced at that moment: FULL (50% chance) pays the whole pot to the survivor, HALF (30%) pays half to the survivor and half to the treasury, and EMPTY (20%) sends the whole pot to the treasury.

Between eliminations, any survivor can force a decision on everyone else: propose a raise of +20 SQK - always the entry amount, never more. How many raises a game allows depends on its lobby: 1 in QUICK 4, 2 in CLASSIC 9 and 3 in BIG 16. A raise then runs in two phases: 30 seconds to decide - accept or decline only, with silence counting as a decline and decliners eliminated - and then, for the players who accepted, a separate 60-second window to submit and confirm the +20 SQK transaction on Sepolia; an unconfirmed transaction also means elimination. Both windows end early: the decision window closes the moment every alive player has accepted or declined, and the transaction window closes the moment every accepted player's transaction is confirmed (or has failed). Raises grow the pot and the fees in the same 18 / 2 split.

Nothing sent to the treasury is burned or kept. Every 6 hours - every 1,800 Sepolia blocks, at about 12 seconds per block - anyone can trigger the treasury draw: the entire treasury is raffled among every address that entered a game before that draw block (in the current draw period) and airdropped to one winner. A player who loses a game can still win the draw.

Status - read this first

The contract is **deployed and live on Sepolia**. `SqueekTiles: 0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212` (deploy transaction `0x7072a5b6824f1657e93d8550c7dd9231e9e9a58098a567539632af8237c0908d`; view it on Sepolia Etherscan at `https://sepolia.etherscan.io/address/0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212`). In Real Mode, players connect MetaMask on Sepolia and real Sepolia SQK moves through this contract. Practice mode also remains fully playable, with the complete rules, bot opponents and a demo ledger. The live game link and the project's own domain will be filled in at submission time - the game-site link is therefore left as **to be added** in this paper. This is a testnet deployment only: SQK on Sepolia has no real-world value, and the contract is unaudited (see Section 12). Section 13 records the deployment and the remaining submission steps.

SECTION 1

Why Squeek Tiles exists

Most on-chain elimination and lottery games ask the player to trust three things they cannot check: that the draw was really random, that the prize split was decided fairly, and that the operator cannot quietly change the rules mid-game. Squeek Tiles is designed so that none of those three trusts is needed.

First, the most consequential outcome - whether the final tile pays FULL, HALF or EMPTY - is not decided at the end of the game, when the operator knows who won. It is decided at the start, from a block that already exists, by a formula printed in this paper, on the game board and in the contract. Anyone can recompute it.

Second, the economics cannot drift. The entry is always 20 SQK, the split is always 18 to the prize and 2 to the creator, and a raise is always another 20 SQK - never a custom amount. A wealthy player cannot invent a 1,000 SQK raise to push everyone else out. Pressure exists, but it is capped by the rules, not by wallets.

Third, losing is not the end of the value. HALF and EMPTY outcomes feed a treasury that returns to the current draw period's players through a treasury draw every 6 hours. The game therefore has two ways to win - survive the board, or be drawn from the period - and both are defined in the contract, not on a server.

Design principles

- **Rules in code, not on a server.** Entry, splits, tiers, raise caps and the 6-hour treasury draw all live in `SqueekTiles.sol`.
- **Decided early, revealed late.** The final tile is sealed at game start and only revealed when the survivor reaches it. The tension is preserved; the tampering window is closed.
- **No owner controls.** The contract as written has no pause, no withdrawal by an owner and no way to re-roll a tier.
- **Reviewable without money.** Practice mode runs the identical rules with fake tokens, so Genesis reviewers and new players can play a full game before any wallet is involved.

SECTION 2

How it uses Squeekchain

Squeek Tiles is built for Squeekchain and uses it in three distinct ways - as the money layer, as the clock and anchor for the final tile, and as the public record of the game.

1. Money layer - SQK on Sepolia. Entry and raises are SQK token transfers through the deployed SqueekTiles contract at `0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212` on Sepolia. The SQK token is `0xd78a3956cF63164d1Fa1C622ad3c66623162Fa1e` (18 decimals). A player connects MetaMask on Sepolia, approves the contract to spend SQK, and then joins a lobby - the site resolves the underlying `joinLobby()` call automatically, with no manual game ID to enter (see Section 4). That transaction is the entry: 20 SQK leaves the player's wallet, 18 SQK stays in the contract as prize and 2 SQK is transferred to the creator address `0x8a53644c0CF45EFc0247E8e01d24607b870793C4` in the same call. There is no off-chain balance that the game asks anyone to trust.

2. The final-tile seed - the block at game start. The moment a lobby fills, the contract reads the current block number and applies the published formula:

```
seed = (blockNumber x 2654435761 + 97) mod 100

0-49 = FULL (50%)  ·  50-79 = HALF (30%)  ·  80-99 = EMPTY (20%)
```

The demo board polls the live Squeekchain status API for the current block height and shows the same formula next to it, so the link between the chain and the game is visible while playing, not hidden in code. The multiplier 2654435761 is the well-known Knuth multiplicative hash constant; it spreads consecutive block numbers evenly across 0-99, so neighbouring games do not cluster on the same tier.

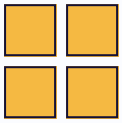
3. Public record. Every meaningful event - `Joined`, `GameStarted` (with seed block and seed), `Eliminated`, `RaiseProposed`, `GameFinished` and `TreasuryDrawn` - is emitted as a contract event. The history of a game can be reconstructed from the chain alone. The block count is also the treasury clock: a draw becomes available every 1,800 blocks (about 6 hours on Sepolia), so the airdrop schedule is set by the chain, not by a server clock.

> **A clarification on randomness**
>
> Elimination itself is a random draw *among the entered addresses* - it does not use live hamster telemetry (RPM, distance or speed) as its input. The Squeekchain link for elimination is the on-chain draw in the contract; the Squeekchain link for the prize tier is the block-seed formula above. Section 12 states the limits of the draw honestly.

SECTION 3

The square rule and the three lobbies

The board is the identity of the game. Every player is one tile, and all tiles together must form one perfect square. That single constraint decides which games can exist at all: 5, 7, 10 or 12 players can never start a game, because no square can be built from them. There is no partial board and no game that starts early with the wrong count.



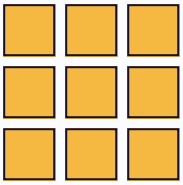
QUICK 4

2 x 2 board

Starting pot 72 SQK

3 eliminations to a winner

Fastest game



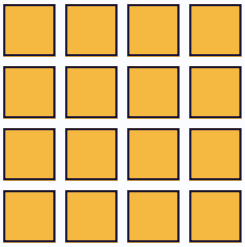
CLASSIC 9

3 x 3 board

Starting pot 162 SQK

8 eliminations to a winner

The main lobby



BIG 16

4 x 4 board

Starting pot 288 SQK

15 eliminations to a winner

Largest pot, longest game

Before joining, the player chooses the lobby. The choice is a choice about game length and pot size, not about odds: the entry, the split and the tier odds are identical in all three lobbies. Only the raise cap differs, scaled to the lobby: 1 in QUICK 4, 2 in CLASSIC 9 and 3 in BIG 16. A 16-player game simply runs more rounds and accumulates a larger pot.

Lobby	Board	Entries collected	To prize (18 each)	Creator fees (2 each)	Elimination rounds	Max raises
QUICK 4	2 x 2	80 SQK	72 SQK	8 SQK	3	1
CLASSIC 9	3 x 3	180 SQK	162 SQK	18 SQK	8	2
BIG 16	4 x 4	320 SQK	288 SQK	32 SQK	15	3

Tile positions are shuffled when the game starts, so joining first gives no favoured tile. Position carries no advantage: elimination draws from the list of alive addresses, not from board coordinates.

Who can join

One wallet address holds at most one tile in a given game - the contract rejects a second entry from the same address (`already entered`). The same address may of course play in different games, and every game it enters adds it to the current draw period's treasury draw list (once per address per period), for the draw at the next 1,800-block boundary.

Why the square matters for fairness

Because a game can only start at exactly 4, 9 or 16, nobody can be pushed into a game whose size, pot or round count changed after they paid. The lobby a player chose is the game they get.

SECTION 4

The full process, step by step

CONNECT -> APPROVE SQK -> JOIN LOBBY -> BOARD FILLS -> SPIN + ELIMINATE ->
RAISE? (cap by lobby) -> FINAL TILE -> PAYOUT / TREASURY -> TREASURY DRAW (6H)

1 **Connect a wallet.** In Real Mode the player connects MetaMask on Sepolia; the site checks the network and shows the wallet's SQK balance. In practice mode this step is simulated and a practice wallet with 1,000 fake SQK is created instead, with a faucet for +1,000 more.

2 **Approve SQK.** Before the first entry, the player approves the SqueekTiles contract to spend SQK from their wallet - the standard ERC-20 allowance step, without which the entry transfer cannot happen. Raise payments need allowance too.

3 **Choose a lobby size and join - lobby selection is automatic.** The player picks only a lobby size - QUICK 4, CLASSIC 9 or BIG 16 - and presses Join. There is no manual game ID to enter: the site finds an open lobby of that size and joins it, or creates a new lobby if none is open, and then calls `joinLobby()` for that game. The game ID is shown for information only. The contract takes exactly 20 SQK: 18 SQK is added to that game's pot and held in the contract, 2 SQK is transferred immediately to the creator address. The player's address is recorded, marked alive, and added to the current draw period's entrant list for the next treasury draw. Their hamster tile appears on the board.

4 **The lobby fills and the game starts - automatically.** While the lobby is still filling, the site shows live how many players have joined and how many more are needed - for example, 2 of 4 joined, 2 more to start - so a waiting player always knows the state of their lobby. The join that completes the square (the 4th, 9th or 16th entry) triggers `_start()` in the same transaction, and the game starts automatically with no further button press. The contract records the current block as the seed block, computes the seed and tier by the published formula, and sets the first spin to become available after the spin interval. From this moment the tier is locked. It is shown in the game as sealed - its value exists, but it is only revealed at the end.

5 **The wheel spins and one tile falls.** After the spin interval (10 seconds - see Section 14), anyone may call `spinAndEliminate(gameId, round)`. One address is drawn at random from the addresses still alive and is eliminated: its tile greys out and leaves the alive set. The draw cannot be called early - the contract rejects it while the wheel is still spinning.

The full process, continued

6 **Optionally, a raise interrupts the rhythm.** After an elimination, any survivor may propose a raise instead of letting the next spin proceed. The full raise flow is Section 7: everyone alive first gets 30 seconds to accept or decline (decline or silence = eliminated on that tile; the window ends early once everyone has answered), and the acceptors then get a separate 60 seconds to confirm their +20 SQK transaction - an unconfirmed transaction is also an elimination, and this window too ends early once every accepted transaction is confirmed or has failed. A raise can remove several tiles at once and shrink the board faster than spins alone (full detail in Section 7).

7 **One tile remains.** As soon as exactly one address is alive - whether by spins, by a raise settlement, or by both - the game finishes in that same call. There is no separate claim step for the main prize.

8 **The final tile opens.** The locked tier is revealed and the pot is split exactly as the tier dictates: FULL pays 100% to the survivor; HALF pays 50% to the survivor and 50% to the treasury; EMPTY pays 0% to the survivor and 100% to the treasury. The survivor's share is transferred by the contract immediately.

9 **The treasury draw closes the period.** Every 1,800 Sepolia blocks (about 6 hours, at roughly 12 seconds per block) a draw becomes available. Anyone may then call `treasuryDraw()` when the treasury is non-empty and at least one address entered a game before that draw block. One address is drawn at random from the current period's entrant list and the entire treasury is transferred to it; the list then resets for the next 1,800-block period.

What the player sees

Every tile with its short address, the spinning wheel with its countdown bar, the locked pot in the final-tile panel, the seed block and value, and a running log of every entry, elimination, raise and payout. Eliminated tiles stay visible, greyed out.

SECTION 5

Tokenomics - where every SQK goes

Squeek Tiles has one token (SQK), one entry price and one split. There are no other charges, no hidden rake and no second currency. Every payment - the entry and each matched raise - is exactly 20 SQK, split in exactly the same way:

Destination	Per 20 SQK payment	Share	When it moves
Final-tile prize pot (held in contract)	18 SQK	90%	Immediately on each payment
Creator fee (to builder address)	2 SQK	10%	Immediately on each payment
Treasury	0 SQK directly	-	Receives only from HALF / EMPTY final tiles

The creator fee is therefore not taken from the winner at the end; it is taken transparently at the door, per payment, and the pot is always the clean 18-per-payment figure shown on the final tile. The fee recipient is the immutable `creator` address set once in the contract constructor - in the deployed contract it is `0x8a53644c0CF45EFc0247E8e01d24607b870793C4` - and it cannot be changed.

The three money pools

- **The pot** belongs to one game. It starts at 18 x lobby size and grows by 18 per confirmed raise payment. It is emptied completely when the game finishes - nothing carries over between games.
- **Creator fees** leave the contract immediately and never return: 2 x (entries + confirmed raise payments) per game.
- **The treasury** is shared across all games. It only grows when a final tile lands HALF (half a pot) or EMPTY (a whole pot), and it is emptied completely by each 6-hour treasury draw.

Conservation check

For any finished game, the following must hold exactly, and any reader can verify it from the contract events:

```
total paid in = pot paid to winner + pot sent to treasury + creator fees
where total paid in = 20 x (entries + confirmed raises), and pot = 18 x (entries + confirmed raises)
```

No SQK is created, burned or left unaccounted inside the contract by the game logic. A player who declines a raise loses only the payments already made - those stay in the pot and fees, as if eliminated by the wheel. There are no refunds, by design: a refundable entry would let a player threaten to leave at no cost and break the raise mechanic.

Test-token warning

SQK in this build is the Sepolia test deployment. Sepolia tokens have no real-world value and exist so the game can be played, reviewed and audited safely. Any future move to a live network would be a separate deployment and decision.

SECTION 6

Worked examples with exact numbers

Example A - QUICK 4, no raises, FULL tile

Step	Amount	Running pot
4 entries x 20 SQK (18 prize + 2 fee each)	80 SQK in; 8 SQK fees out	72 SQK
3 spins eliminate 3 tiles	-	72 SQK
Final tile: FULL	72 SQK to survivor, 0 to treasury	0

Check: 72 (winner) + 0 (treasury) + 8 (fees) = 80 paid in. The survivor paid 20 and receives 72.

Example B - QUICK 4, one raise confirmed by all, HALF tile (the tested game)

This is the exact game used to test the built demo, step for step:

Step	Amount	Running pot
4 entries x 20 SQK	80 SQK in; 8 SQK fees out	72 SQK
First spin eliminates 1 tile (3 remain)	-	72 SQK
A survivor proposes a raise (QUICK 4 allows exactly 1). In the 30-second decision phase all 3 alive players accept - so the window ends early - and in the 60-second transaction phase all 3 confirm +20 SQK (18 prize + 2 fee each), so that window ends early too.	60 SQK in; 6 SQK fees out; +54 to pot	126 SQK
Spins continue until 1 tile remains	-	126 SQK
Final tile: HALF	**63 SQK to survivor, 63 SQK to treasury**	0
Treasury draw (at the next 1,800-block boundary, about 6 hours)	Treasury, including the 63 SQK, drawn among all addresses that entered before that draw block	-

Check: 63 (winner) + 63 (treasury) + 14 (fees: 8 entry + 6 raise) = 140 = 80 entry + 60 raise paid in. Note the survivor in this game paid 40 SQK in total (entry + raise) and received 63 - a raise that everyone confirms grows the pot for whoever survives, it does not guarantee a profit.

Worked examples, continued

Example C - CLASSIC 9, two raises, one player declines the second

Step	Payments	Running pot
9 entries	180 in; 18 fees	162 SQK
Raise 1: proposed with 6 alive; all 6 accept (30 s decision, ends early once all answered) and confirm (60 s transaction, ends early once all confirmed)	120 in; 12 fees	270 SQK
Spins reduce the board to 4 alive	-	270 SQK
Raise 2 (the CLASSIC 9 cap of 2 is now reached): proposed with 4 alive; 3 accept and confirm, 1 does not answer in the 30-second decision phase	60 in; 6 fees	324 SQK
Settlement: the silent player counts as a decline and is eliminated on that tile; the 3 acceptors proceed and confirm in the transaction phase	-	324 SQK

The game then continues with 3 alive and 324 SQK locked. If the final tile is EMPTY, all 324 SQK go to the treasury and the survivor of the board receives nothing from the pot - their remaining chance in that period is the 6-hour treasury draw, which every entrant before the draw block shares.

Example D - BIG 16 at maximum raises

The largest possible pot in the current rules: 16 entries (pot 288) plus 3 raises - the BIG 16 cap - confirmed by a shrinking field. If, for illustration, all 16 were still alive for raise 1, 12 for raise 2 and 8 for raise 3, the raise payments add (16 + 12 + 8) x 18 = 648 SQK, for a final pot of 936 SQK and total fees of 2 x (16 + 36) = 104 SQK on 1,040 SQK paid in. In practice fields shrink faster, so real pots land well below this ceiling - the ceiling exists to show that even the maximum case is bounded and computable in advance.

SECTION 7

The raise system - pressure with a hard cap

The raise is the game's only player-controlled weapon, and it is deliberately narrow. After any elimination, while the game is in the Playing phase, any surviving player may call `proposeRaise()`. Proposing counts as the proposer's own acceptance. The raise then runs in two separate phases with two separate clocks - deciding is not paying, and the game never confuses the two:

- **Fixed amount.** A raise is always +20 SQK - the entry amount. There is no parameter for a larger raise, in the interface or in the contract. Nobody can name a price others cannot afford.
- **Hard cap, scaled to the lobby.** At most **1 raise in QUICK 4, 2 raises in CLASSIC 9 and 3 raises in BIG 16** (`maxRaises(size)` = 1 / 2 / 3), counted whether they succeed or fail. Any proposal beyond the lobby's cap is rejected by the contract. Three raises would dominate a short 4-player game; one raise is the right pressure for it, while the longer 9- and 16-player games can absorb two and three.
- **Phase 1 - Decision (30 seconds).** Every other alive player signals **accept** or **decline**. This phase is decision only: no SQK moves and no payment transaction is sent yet. The board shows live who has accepted, who has declined and who has not answered. The phase ends early the moment every alive player has answered.
- **Silence is a decline.** When the 30 seconds close, every alive player who declined or never answered is eliminated - all of them, on that tile, at once - by `settleRaise()`. Their tiles leave the board. If exactly one player remains, the game finishes immediately and Phase 2 never opens.
- **Phase 2 - Transaction (60 seconds).** The players who accepted now actually pay: each one submits the +20 SQK raise transaction (the `payRaise()` call, split 18 to the pot and 2 as fee, with the proposer's payment taken in this same phase) and that transaction must be **confirmed on Sepolia within 60 seconds**. Accepting is a promise; confirming is the payment. Anyone whose transaction is not confirmed when the 60 seconds close is eliminated at settlement, exactly as if they had declined - a slow wallet, a rejected signature or an insufficient balance all count the same way. Only confirmed payments add to the pot. **Early end:** this window does not have to run its full 60 seconds - it ends immediately once every accepted player's transaction is confirmed, or has failed and that player is eliminated. The same early-end rule applies in Phase 1: the 30-second decision window ends immediately once every alive player has accepted or declined.
- **Declining is a real choice.** A player who has already paid 20 or 40 SQK may judge the pot, the field and the sealed tier risk and walk away. Declining loses past payments but stops future ones. That judgement - not wallet size - is the skill the raise tests.

Why the cap matters

Without a cap, a raise war is a wealth contest: the richest wallet proposes again and again until everyone else folds. With fixed raises and a cap scaled to the lobby, the maximum extra exposure is known before joining: QUICK 4 - at most 1 raise, 20 SQK on top of the 20 SQK entry, 40 SQK worst case; CLASSIC 9 - at most 2 raises, 40 SQK extra, 60 SQK worst case; BIG 16 - at most 3 raises, 60 SQK extra, 80 SQK worst case. A player can decide in advance whether a game is affordable at its maximum, not just at its entry.

SECTION 8

The final tile: FULL, HALF, EMPTY

Every game ends at the same place: a single golden final tile holding the whole pot. What the survivor finds inside was decided before the first spin, by the block-seed formula of Section 2, and is revealed only when the game finishes.

Tier	Seed	Chance	Survivor receives	Treasury receives
FULL	0-49	50%	100% of pot	0
HALF	50-79	30%	50% of pot	50% of pot
EMPTY	80-99	20%	0	100% of pot

Three properties make this fair rather than theatrical. **Pre-commitment:** the seed block, the seed and the tier are fixed in `_start()` and the seed block and seed are emitted in the `GameStarted` event - the commitment is public from the first second. **Immutability:** no function in the contract writes to the seed or tier after start; there is nothing to call that changes them. **Recomputability:** given the seed block number, any reader can apply (block x 2654435761 + 97) mod 100 on paper and confirm the tier the contract used. Odd pots split by integer division in the contract's favour of the treasury by at most a fraction of a token's smallest unit - with 18-decimal SQK this remainder is negligible and stays in the treasury side of the split, never with an operator.

The expected payout to a survivor, before considering raises, is therefore 0.5 x pot + 0.3 x (pot / 2) + 0.2 x 0 = 65% of the pot on average - and the remaining 35% on average is not lost to the house, it is recycled to the current draw period's players through Section 9.

SECTION 9

Treasury and the 6-hour airdrop

The treasury is the game's memory of unlucky endings. It has exactly two sources - the treasury half of a HALF tile and the whole pot of an EMPTY tile - and exactly one destination: one randomly drawn entrant from the current draw period.

- **Who is in the draw:** every distinct address that entered any game before the draw block, in the current draw period - whether they won, were eliminated in the first spin, declined a raise or survived to an EMPTY tile. Entry is recorded once per address per period, so entering five games in one period does not give five tickets - it gives one ticket and five games of play.
- **When it runs:** the draw is scheduled by block count, not by a wall clock. Sepolia produces a block roughly every 12 seconds, so a fixed interval of **1,800 blocks is about 6 hours** (1,800 x 12 seconds = 21,600 seconds = 6 hours). Once 1,800 blocks have passed since the last draw block, anyone may call `treasuryDraw()` - no operator or owner is needed. If the treasury is non-empty and at least one address entered before that draw block, the function draws one address from the period's entrant list and transfers the entire treasury to it, emitting `TreasuryDrawn`.
- **Period rollover:** the contract tracks the last draw block (`lastDrawBlock`) and the interval `DRAW_INTERVAL = 1800 blocks`. After a draw, the entrant list is cleared and the next period runs to the next 1,800-block boundary, so one period's players are never drawn against another period's treasury.

> **The design intent**
>
> EMPTY is the harshest outcome in the game - a player survives 3, 8 or 15 eliminations and receives nothing from the pot. The treasury draw is the counterweight: that same pot returns within about 6 hours to the pool of people who played in that period, possibly including the EMPTY survivor themselves. Value circulates among players instead of accumulating anywhere.

What the treasury is not

Creator fees never enter the treasury and are never drawn - they are paid out at entry and are final. The treasury also has no owner withdrawal function: the only transfer out of it in the contract is the treasury draw itself.

SECTION 10

Practice mode vs on-chain mode

	Practice mode (live today)	Real Mode - on-chain (live on Sepolia)
Ledger	In-page demo ledger, session only	Deployed SqueekTiles contract on Sepolia
Tokens	Fake SQK - 1,000 on connect, faucet +1,000	Real Sepolia SQK via wallet approval + transfer
Opponents	Bot players with random addresses	Other wallets that joined the same lobby
Wallet	Optional; simulated if no extension is found	MetaMask on Sepolia required; entry is a signed transaction
Lobby	Filled instantly by bots	Automatic: pick a size and the site finds or creates an open lobby; live joined / needed count while it fills; starts automatically when full
Rules	**Identical** - same lobbies, split, tiers, formula, raise caps and draw	
Block seed	Live Squeekchain block height when reachable, labelled simulated feed otherwise	The actual block number inside the start transaction

Practice mode exists for three reasons: Genesis reviewers must be able to judge the game without owning SQK; new players should learn the raise decision with fake money before real test tokens; and the rules themselves should be provable in the browser before anyone is asked to trust a contract. Practice mode is unchanged by the deployment and never pretends to be the chain - its wallet panel labels it as practice, and its chain bar labels a simulated block feed as simulated when the live API cannot be reached. Real Mode is the on-chain counterpart: the player connects MetaMask on Sepolia, approves SQK, picks a lobby size and joins - lobby selection is automatic, with no manual game ID, and while a lobby fills the site shows live how many players have joined and how many more are needed, starting the game automatically when the lobby is full. Both modes follow the same official timings as Section 14 - a 10-second spin with no skip option, and the two-phase raise (30 seconds to decide, then 60 seconds to confirm, both ending early once everyone has responded or confirmed). The timings in the browser are presentation of the same contract rules, not a separate ruleset.

SECTION 11

Smart contract specification

Contract: `SqueekTiles.sol` - Solidity ^0.8.24, MIT licence, single contract plus a minimal `IERC20` interface. The constructor takes two immutable addresses only: the SQK token and the creator (fee recipient). **Deployed on Sepolia** at `0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212`, with SQK token `0xd78a3956cF63164d1Fa1C622ad3c66623162Fa1e` (18 decimals) and creator `0x8a53644c0CF45EFc0247E8e01d24607b870793C4`. Deploy transaction and Etherscan link: Section 13.

Constant	Value	Meaning
`ENTRY`	20 SQK	Entry and raise payment - always the same
`TO_PRIZE / FEE`	18 / 2 SQK	The split applied to every payment
`SPIN_TIME`	10 seconds	Wheel spin per round. There is no skip function.
`RAISE_DECISION_TIME`	30 seconds	Phase 1 of a raise: accept or decline only - no transaction yet. Ends early once every alive player has answered.
`RAISE_TX_TIME`	60 seconds	Phase 2 of a raise: submit and confirm the +20 SQK transaction. Ends early once all accepted transactions resolve.
`MAX_RAISES (by lobby)`	1 / 2 / 3	Raise cap per game: QUICK 4 = 1, CLASSIC 9 = 2, BIG 16 = 3
`DRAW_INTERVAL`	1,800 blocks	Treasury draw interval on Sepolia - about 6 hours at roughly 12 seconds per block

Phase	What happens in it
`Filling`	Entries accepted until the lobby size is reached; the filling join starts the game.
`Playing`	Spins and eliminations run; raises may be proposed.
`RaiseDecision`	Phase 1 of a raise: 30 seconds to accept or decline; no funds move; spins are paused.
`RaisePayment`	Phase 2 of a raise: 60 seconds for acceptors to submit and confirm the +20 SQK transaction; unconfirmed = eliminated. Ends early once all accepted transactions are resolved.
`Finished`	Tier revealed, pot split and paid; the game is read-only history.

Player functions: `createGame(size)` - size must be 4, 9 or 16; `joinLobby(gameId)` - approve-then-join entry; `proposeRaise(gameId)` ; `acceptRaise(gameId)` / decline by not answering in Phase 1; `payRaise(gameId)` in Phase 2. In Real Mode the player never types a game ID: the site calls `createGame()` / `joinLobby()` itself to find or create an open lobby of the chosen size. **Keeper functions** (callable by anyone, no privilege): `spinAndEliminate(gameId, round)` , `settleRaise(gameId)` , `treasuryDraw()` . Making progression callable by anyone means a game can never be stalled by an absent operator - any player can advance their own game once its timer allows.

SECTION 12

Fairness, security and honest limitations

What is protected

- **Tier integrity:** set once from a past block at start, emitted publicly, never writable again.
- **Double entry:** rejected per address per game; raise double-payment is likewise rejected.
- **Timer integrity:** eliminations before `spinReadyAt` and raise settlements before their phase deadline revert - unless every player has already responded (decision phase) or every accepted transaction is already confirmed or failed (transaction phase), in which case early settlement is allowed and the window ends immediately.
- **Fund safety:** payouts use checked ERC-20 transfers (`require(transfer)`); a failed payout transfer reverts the finish rather than silently losing the pot.

What is not claimed

- **The elimination draw is not a VRF.** It derives from `keccak256(blockhash, gameId, round, seed)`. On a test network with test tokens this is proportionate, but blockhash-based draws are in principle influenceable by block producers. A production version handling real value should replace the draw with a verifiable random function (VRF) oracle. This paper does not pretend otherwise.
- **One address is not one person.** Nothing stops one person from playing through several wallets, in several games or - across different games in the same draw period - holding several draw entries only if the addresses differ, since the draw list is per address per period. The per-game one-tile rule still limits any single game.
- **No audit yet.** The contract is published for review and mirrors a tested demo, but it has not had an independent security audit. Sepolia-first deployment is precisely so that review happens where mistakes cost nothing.

SECTION 13

Deployment record and submission

This section records what was deployed, with the exact addresses, and lists only the steps that remain before submission.

1. **Deployed on Sepolia - done.** SqueekTiles is live at `0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212`. Deploy transaction: `0x7072a5b6824f1657e93d8550c7dd9231e9e9a58098a567539632af8237c0908d`. Etherscan: `https://sepolia.etherscan.io/address/0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212`.

2. **Constructor values - fixed at deployment.** SQK token (`sqk`): `0xd78a3956cF63164d1Fa1C622ad3c66623162Fa1e` (18 decimals). Creator / fee recipient (`creator`): `0x8a53644c0CF45EFc0247E8e01d24607b870793C4`, receiving every 2 SQK fee. Both are immutable and cannot be rotated.

3. **Final rules - as deployed.** `ENTRY` 20 SQK split 18 / 2, `SPIN_TIME` 10 seconds with no skip function, the two-phase raise - `RAISE_DECISION_TIME` 30 seconds plus `RAISE_TX_TIME` 60 seconds, both ending early once everyone has responded or confirmed - raise caps of 1 / 2 / 3 by lobby, and `DRAW_INTERVAL` 1,800 blocks (about 6 hours) for `treasuryDraw()`. Changing any of this would require a new deployment.

4. **Front end wired - done.** Real Mode connects MetaMask on Sepolia, shows the SQK balance, and runs Approve then Join against the deployed contract, with per-transaction status and Etherscan links. Lobby selection is automatic - the player picks a lobby size only, the site finds or creates an open lobby, shows live how many players have joined and how many more are needed while it fills, and starts the game automatically when full. Practice mode stays available as a separate, clearly labelled mode.

5. **Remaining: one full real test game.** Run a complete 4-player game end-to-end with real Sepolia SQK - entries, eliminations, a raise if proposed, the final-tile payout and, when due, a treasury draw - before submitting.

6. **Remaining: domain and live link.** The live game link and the project's own domain will be filled in at submission time; in this paper the game-site link is left as **to be added**. Direct Submission requires the site on the project's own domain, with the DNS record and page tag provided after submitting used to prove ownership.

7. **Remaining: submit to Genesis** with the live game link on the project's own domain, the source repository, the deployed contract address above and this whitepaper. The Genesis form's explanation of how the project uses Squeekchain is covered in Section 2 of this paper; the form itself is completed by the builder at submission time. The deployment remains testnet-only and the contract unaudited (see Section 12).

SECTION 14

Timing: the final settings

The official timings are set: the wheel spins for 10 seconds - with no skip option for anyone - and the raise is a two-phase decision: 30 seconds to accept or decline, then a separate 60 seconds to submit and confirm the +20 SQK transaction on Sepolia. Both raise windows end early: the decision window closes the moment every alive player has accepted or declined, and the transaction window closes the moment every accepted player's transaction is confirmed (or has failed). An earlier draft lengthened the wheel to 15 seconds; play-testing showed the original 10 seconds keeps the game moving, and the real fix for feeling rushed was never a longer wheel - it was separating the human decision from the blockchain confirmation, and letting finished windows close instead of idling.

First, separate the three clocks

Clock	What it is	Can the game change it?
Squeekchain / Sepolia blocks	Sepolia produces a block roughly every 12 seconds. That block time sets the treasury rhythm: 1,800 blocks is about 6 hours. A game round is *not* a chain round.	No - and it does not need to. The chain supplies the seed block at game start and the 1,800-block draw schedule, not a per-spin heartbeat.
Demo timers	The 10-second spin animation, the 30-second raise decision countdown and the 60-second raise transaction countdown, as rendered in the browser in practice mode.	Yes - front-end presentation, freely tunable, as long as it mirrors the contract.
Contract parameters	`SPIN_TIME = 10s`, `RAISE_DECISION_TIME = 30s` and `RAISE_TX_TIME = 60s` enforced on-chain, with early settlement once everyone has responded or confirmed.	No - fixed in the deployed contract. Changing them would require a new deployment.

On-chain there is a further reality the demo hides: a timer expiring does not make anything happen by itself. A spin only occurs when someone sends the `spinAndEliminate` transaction, and a wallet signature plus a Sepolia block confirmation typically take longer than the nominal spin timer. Real on-chain games will therefore always run a little slower than the practice demo at the same settings - one more reason the wheel stays at 10 seconds rather than being inflated. There is deliberately **no Skip Spin function**: in the first draft a skip control raised an obvious fairness question - if one player could end the 10-second spin for everyone, the spin would not be the same event for all players. The spin now always runs its full 10 seconds, for everyone, every round.

The official timings in detail

Moment	Earlier draft	Official	Why
Wheel spin FINAL	15 s in the previous revision	**10 s - no skip**	Watching time. Ten seconds is enough to read the board, the alive count and the last elimination while the wheel turns, and it keeps a 16-player game (15 spins = 150 s of pure spinning) inside two and a half minutes. No player, button or function can shorten it for anyone.
Elimination reveal pause ADD	0 s	**3 s pause** (front-end)	A 3-second beat after each elimination lets players register who fell and check the alive count before the next spin. Costs little: +45 s over a full BIG 16 game.
Raise - Phase 1: decision FINAL	20 s (single window, decision + payment together)	**30 s - accept or decline only - ends early**	Deciding time. Thirty seconds is enough to read who proposed, count the alive players, check the pot and your balance, and make the call - no wallet opens and no transaction is sent in this phase. **The window ends immediately once every alive player has accepted or declined**; players are never left watching a countdown that has nothing left to decide. Silence or an explicit decline at the deadline eliminates the player.
Raise - Phase 2: transaction NEW	Did not exist - payment had to fit inside the same 20 s	**60 s - submit and confirm on Sepolia - ends early**	Confirmation time. Only the players who accepted enter this phase. Each submits the +20 SQK transaction and it must be confirmed on Sepolia before the 60 seconds end. **The window ends immediately once every accepted player's transaction is confirmed, or has failed and that player is eliminated** - fast confirmations are never punished with a wait. Not confirmed in 60 seconds = eliminated, even after accepting.
Lobby / start countdown ADD	Instant start on fill	**5 s "game starting" countdown** (front-end)	The filling join starts the game in the same transaction on-chain; the countdown is presentation only, giving every player a moment to focus before spin 1 and making the seed-lock moment visible.
Final tile reveal ADD	Instant	**4 s sealed-tile pause** (front-end)	The tier reveal is the payoff of the whole game. Holding the sealed tile for a few seconds before it opens costs nothing and makes FULL / HALF / EMPTY land as a moment, not a flicker.
Treasury draw CHANGED	Once per day	**Every 1,800 blocks - about 6 hours**	A daily wait was too slow for testnet play: pots sent to the treasury would sit for a day. At Sepolia's roughly 12-second block time, 1,800 blocks is about 6 hours - frequent enough that the airdrop is a live part of the game, and anyone can trigger it once the interval has passed.

What the final game looks like - and the appendix

What the official timings do to a full game

Lobby	Spins only (10 s wheel)	With reveal pauses	Plus one full raise (30 s + 60 s, at most)
QUICK 4 (3 spins)	~30 s	~40 s	~2 min at most - usually less, because both raise windows end early once everyone has responded or confirmed
CLASSIC 9 (8 spins)	~1 min 20 s	~1 min 45 s	~3-5 min with its maximum of 2 raises
BIG 16 (15 spins)	~2 min 30 s	~3 min 15 s	~5-8 min with its maximum of 3 raises

That is the right shape for this game: a quick lobby that still fits in a coffee break's first minutes, and a big lobby that feels like a short event rather than a wait. The official timings, in one line as fixed in the deployed contract: **`SPIN_TIME` = 10 seconds, no skip; raise = 30 seconds to decide + 60 seconds to confirm, both ending early; treasury draw every 1,800 blocks (about 6 hours); plus the 5 s / 3 s / 4 s presentation pauses in the front end.** Because of the early-end rule, a full raise costs at most 90 seconds and usually much less. If play-testing still feels rushed, the honest lever is the front-end pauses - not inflating the wheel.

Appendix - Glossary

Tile: one player's position on the board, shown as a hamster tile with their short address. **Final tile:** the golden tile holding the pot; its tier is sealed at game start. **Seed:** the 0-99 number produced by the block formula; it determines the tier. **Raise:** a forced +20 SQK decision proposed by a survivor; the number allowed per game depends on the lobby (1 / 2 / 3). **Draw period:** the 1,800-block interval (about 6 hours) between treasury draws. **Treasury:** the shared pool fed by HALF / EMPTY tiles and emptied by the treasury draw every 1,800 blocks. **Practice SQK:** fake tokens in the demo ledger, with no value and no chain transaction.

Squeek Tiles Whitepaper v1.0 - Real Mode is live on Sepolia (contract 0x7042Fa3563706eAcc8688AF9C3bBcf4Ae1C6F212) and Practice mode is live. Live game link and project domain: to be added at submission. SQK on Sepolia is a test token with no real-world value. Testnet only; contract unaudited.